Improving Accelerator Control using Hybrid Bayesian Optimization and Reinforcement learning Approach

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Introduction

The Continuous Electron Beam Accelerator Facility (CEBAF) serves to research quarks, gluons, and the structure of protons and neutrons. This proverbial microscope requires a lot of computational and manpower. The object of my research project was to streamline the processes by which experiments are ran via simulation. Proper use of CEBAF requires careful balance between selection of cavity settings while maintaining performance and operational constraints. The number of potential parameters is quite large, which means this is a high-dimension optimization problem. Two existing toolkits were provided Xopt and SoCT. Xopt is a bayesian optimizer that can narrow the problem space with a small amount of iterations, but as dimensionality increases it becomes costly and time consuming to run. SoCT provides a gymnasium based environment where agents can learn a control policy and adapt it as conditions change. Both strategies are complimentary in nature - so the question is: can the information found by Xopt be reused by SoCT to prevent the agent from starting from scratch.

Methods

Initial work revolved around developing and testing a shared workflow between Xopt Bayesian Optimization and the Scientific Optimization Control Toolkit (SOCT)'s reinforcement learning framework. All experiments were conducted on existing simulated CEBAF environments, with an initial baseline of the 8-Dimensional cavity problem which later was scaled to 32-dimensional environments.

Phase 1: Xopt and CEBAF integration

In order to allow Xopt and SoCT to speak the same language, a gymnasium to Xopt interface was developed which allowed Xopt to directly evaluate potential cavity settings within the same CEBAF simulation environment used by SoCT. Typically, Xopt sets candidate settings as optimization variables and then evaluates each configuration based on the environment reward. The CEBAF delta-action environment however expect changes to be relative to the current state - this is counter to Xopt's absolute settings. To address this each Xopt target was converted into an action by subtracting the current state from the proposed target state. To give the learning agent more to work with, observation, reward, termination status, and other environment information were recorded after every evaluations. The complete evaluation history and best state were then exported for later use by the RL framework.

Phase 2: Xopt to SoCT handoff

The first integration bridge treated Xopt as the initializing source for SoCT. The toolkit was rewritten to accept Xopt's best found state as the environment's reset state - This allowed reinforcement-learning episodes to start in an already optimized region instead of the default problem space. All Xopt evaluation were also translated to match TD3's state, action, reward, next-state and terminated transitions language. Additionally the translated evaluations were inserted into the replay buffer right before training. This means TD3 had access to examples of useful and failed accelerator configurations before it started collecting its own data.

Phase 3: Reinforcement Learning Around Xopt Solution

The early handoff iterations showed that jump-starting SoCT with Xopt best state did not guarantee that TD3 wouldn't toss it during exploration. The SoCT configurations were changed to prioritize local state refinement around the Xopt best found solution. Delta actions were used to force the agent to learn changes relative to its current point. Additionally exploration magnitude and learning behavior parameters were constrained to reduce movement away from the optimized area. Training and inference rewards were then inspected separately to evaluate if the learned policy was able to retain to improve on the Xopt starting point.

Phase 4: Hybrid Xopt guided Exploration

The final iteration of the project abandoned the one-time warm start in favor of a more complicated hybrid optimization strategy. Instead of using Xopt before reinforcement learning began, the hybrid approach had Xopt periodically find new optimization target and use them to guide the SoCT exploration. This made it possible for TD3 to keep learning from interacting with the environment and incorporated additional search information from BO to help narrow the problem space. The experiment was standardized to support different CEBAF environment dimensions - meaning the hybrid strategy could scale with the control problems.

Results

The original experiment proved that Xopt can find a strong starting point for the 8 and 32 dimension CEBAF environments. However initializing SoCT with this starting point was not enough to allow SoCT to improve on it. While the TD3 agent did originally stay close to the Xopt baseline, further training steps took the learning policy further away from the optimized area. This is the driver behind the transition from an Xopt based warm start into the replay buffer seeding with reduced exploration, and finally a live-like Xopt assisted exploration. The final hybrid version was successful beyond the 8D experiment, producing promising results in the 32 Dimensional space. This means that the combined Xopt and SoCT interface can operate without needing a fixed amount of cavity controls.

The experiments then compared the live hybrid strategy to the standard TD3 gaussian exploration. The training and inference reward were compared against the best found Xopt state used to initialize each experiment. In the later runs, Xopt guidance did not produce a consistent improvement over standard gaussian exploration but it did keep up with it. The current results serve more as validation that the pipeline works. This will now be a basis for more robust evaluations across multiple seeds, training blocks, and exploration of configurations.

Future work

Future work will center around finding a balance between computing power, time and costs associated with the hybrid method vs the standard TD3 exploration. The current experiments give us an initial comparison point but more experiments across multiple seeds need to be conducted. Some of the higher priority next steps would be an ablation study that compares standard SoCT, Xopt warm-start, replay-buffer seeding, and live Xopt guided exploration to pinpoint which parts of the integration provide the most advantages at the lowest cost. Additionally, experiments on the cadence at which Xopt actions are weaved in to the Reinforcement learning process should be done. This would see adjusting frequency of Xopt query, balancing the Xopt-actions vs the conventional exploration noise, and the amount of BO used before and during the training process. The hybrid approach should be investigated more thoroughly through the lens of larger dimensions. Finding the dimensional break points where the hybrid approach outpaces the standard approach is a good place to start. In a longer term sense, the framework should be extended to accommodate a wider variety of configurations and function on multi-objective optimization. A successful endpoint to this research would be to develop a controller that can take advantage of the search capabilities of BO and the adaptibility of reinforced learning.